Ashmit S Mukherjee

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Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science
Studied at: NYU Abu Dhabi; NYU Paris; NYU New York

Relevant Coursework: Natural Language Processing; Principles of Data Science; Statistics for Social & Behavioral Sciences; Data Structures & Algorithms; Computer Systems Architecture; Operating Systems

Research Interests

I work at the intersection of machine learning and computational social science, with a focus on how artificial agents affect cooperation, coordination, and norm formation. My current work models social dilemmas in mixed human-AI populations and asks when AI-mediated interventions support cooperation, and when they get in the way. My broader work in multilingual NLP and protein representation learning reflects the same methodological interest: how models learn structure from heterogeneous data, and how we test that learning carefully.

Publications

A. Mukherjee (co-first author). "Bio-LoRA: Biological Priors as Inductive Bias for Low-Rank Adaptation." *Under review at EMNLP 2026 Workshop*, 2026.

Research Experience

Research Assistant: Human-AI Collaboration, NYU Abu Dhabi *Feb 2026–Present*

- Developing a simulation-based research platform for studying **AI-mediated coordination in multi-party social dilemmas**, using global supply chains as the domain (advisors: Prof. Hanan Salam, Prof. Benjamin Rosche).
- Built a calibrated game-theoretic environment with five buyers and three suppliers across repeated rounds, combining public goods support, strategic ignorance, threshold remediation, exit versus engagement, and auditing in one domain-grounded model. The main buyer policy uses quantal response equilibrium with a Stage 2 belief solver.
- Reproduced the target inefficient-equilibrium pattern, with about 22-26% welfare loss versus the cooperative ceiling, about 99% supplier defection, and about 7% remediation across a 12-cell calibration sweep with 50 simulations per cell. The backend has 101 passing tests and three completed audit rounds.
- Designing the next phase as an experimental platform comparing algorithmic baselines against human-participant sessions in which AI agents act as advisors, coordinators, monitors, or commitment-support systems.

Research Assistant: eBRAIN Lab, NYU Abu Dhabi *Jan 2026–Present*

- Co-developed **Bio-LoRA**, a parameter-efficient fine-tuning method that injects residue-level biological priors (BLOSUM62, hydrophobicity, Grantham) into LoRA adapters for ESM-2 protein language models on signal peptide prediction.
- Showed BLOSUM62-guided LoRA matches or exceeds full fine-tuning on SignalP6 benchmarks across SP MCC, cleavage-site exactness, and residue-level F1, while training only ~3.6% of parameters and using ~43% less peak GPU memory at the 3B-parameter scale.
- Designed multi-seed evaluation protocols and a one-factor sensitivity screen showing that single-seed improvements often fail to replicate, a methodological finding central to the paper's contribution.
- Manuscript under review (see Publications).

Research Projects

Hinglish Named Entity Recognition Benchmark *Fall 2024*

- Fine-tuned multilingual transformer models, mBERT and XLM-RoBERTa, on the COMI-LINGUA dataset for Hindi-English code-mixed NER.

- Achieved 78% entity-level F1, outperforming zero-shot GPT-4o and LLaMA 3.1 baselines under controlled evaluation.
- Analyzed trade-offs between model scale, task-specific fine-tuning, generalization, and computational cost.

Data Science Salary Prediction Platform (Methodological Study) *Fall 2025*

- Conducted large-scale regression benchmarking using AutoML (PyCaret) across 15+ models on a dataset of 3,000+ job postings.
- Analyzed feature contributions using SHAP to study labor market dynamics across geography, firm size, and skill requirements.
- Tracked experiments with MLflow; reported MAE, RMSE, and R^2 across conditions.

Crime Data Analysis and Prediction *Spring 2025*

- Studied relationships between socioeconomic indicators and violent crime rates using regression modeling on data from 1,994 communities.
- Evaluated model performance via cross-validation ($R^2 = 0.85$) and exploratory statistical analysis; built interactive visualizations for interpretability.

CAMP: Campus Asset Management Platform *Fall 2024*

- Designed and implemented a multi-user system for resource allocation and scheduling under shared constraints, including role-based access and conflict resolution.
- Developed RESTful APIs and automated CI/CD pipelines using Docker and GitHub Actions for reproducible deployment.

Industry Experience

Machine Learning Engineer Intern, Zeek *Jun–Aug 2025*

- Developed and evaluated segmentation models on noisy real-world data, iterating on preprocessing and model selection for production reliability.
- Applied model interpretability techniques to study robustness and bias sensitivity, helping the team decide which model variants to deploy.
- Contributed to the team's evaluation workflow by adding standardized metrics and validation procedures to the model development cycle.

Technical Skills

Languages: Python, C++, C#, JavaScript
ML & NLP: PyTorch, Transformers (Hugging Face), Scikit-Learn, XGBoost, PyCaret, SHAP
Multi-Agent & RL: NumPy, NetworkX, reinforcement learning pipelines
Data Analysis: Pandas, SciPy, Matplotlib, Seaborn, Plotly
Experiment Tracking: MLflow, Git, Docker, GitHub Actions
Other: Streamlit, MongoDB, LaTeX